Week - 07 - Exam Preparation\Lecture_14_code_1.py

```python
class Station :
    def __init__(self, loc, capacity):
        self.loc = loc
        self.capacity = capacity # 20
        self.drones = []        # 10

    def available(self):
        # available_capacity = self.capacity - len(self.drone)
        # return available_capacity
        return self.capacity - len(self.drones)

    # station.dock(self)

    def dock(self, drone):
        # if there is a space available?
        if self.available():
            self.drones.append(drone)
            return True

        else:
            return False

    def recharge(self):
        for drone in self.drones:
            drone = 15
        # TODO check how drone is recharged, and complete this part later

from math import sqrt

class Drone:
    def __init__(self, loc):
        self.loc = loc
        self.range = 15
        self.station = None

    def distance_to(self, loc):
        x1, y1 = self.loc
        x2, y2 = loc

        distance = sqrt( (x2-x1)**2 + (y2-y1)**2 )

        return distance

    def reaches(self, station):
        # dist = self.distance_to(station.loc)
        # if dist <= self.range:
        #     return True
        # return False

        return self.distance_to(station.loc) <= self.range

```

```
    def move(self, loc):
        d = self.distance_to(loc)

        if d <= self.range:
            if self.station is not None:
                self.station = None

            self.range -= d
            self.loc = loc
            return d
        else:
            print("cannot reach location")
            return 0

    def dock(self, station):
        if self.station:
            print("already docked")

        elif not self.reaches(station):
            print("cannot reach station")

        elif station.available() <= 0:
            print("Station full")

        else:
            self.move(station.loc)
            self.station = station
            station.dock(self)
            print("Successfully docked")


s1 = Station((3,2), 1)
s2 = Station((8,6), 2)

d1 = Drone((0,0))
print(d1.move((3,7)))
# 7.615773105863909

print(d1.move((4,4)))
# 3.1622776601683795

print(d1.dock(s2))
# cannot reach station

print(d1.dock(s1))
# successfully docked

d2 = Drone((10,0))
print(d2.move((6,3)))
# 5.0

print(d2.dock(s1))
# station full

```

```
106
107
108 # Time to say goodby to this wonderful class
109 # Please stay happy, and become happier
```